



Lab Report:01

Course Code: ICT - 3104

Course Name: Computer Networks Lab

Submitted to:

Dr. Jesmin Akhter

Professor

Institute of Information Technology

Jahangirnagar University

Submitted By:

MD Khaleduzzaman Fahim

ID:2081

Institute of Information Technology

Jahangirnagar University

Submission Date: 16 July 2025

1.Networking Between Two Computers(using switch):

Network Setup:

Open Cisco Packet Tracer.

From the device list, drag and drop:

- 2 PCs (PC0 and PC1)
- 1 Switch (e.g., Switch0)
- Use copper straight-through cables to connect:
- PC0 to Switch0 (e.g., FastEthernet0/1)
- PC1 to Switch0 (e.g., FastEthernet0/2)

Step-by-Step Configuration:

- Assign IP Addresses to PCs
- Click on PC0 → Desktop tab → IP Configuration
- IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0
- Click on PC1 → Desktop tab → IP Configuration
- IP Address: 192.168.1.3
- Subnet Mask: 255.255.255.0
- (No gateway needed since both PCs are in the same subnet.)

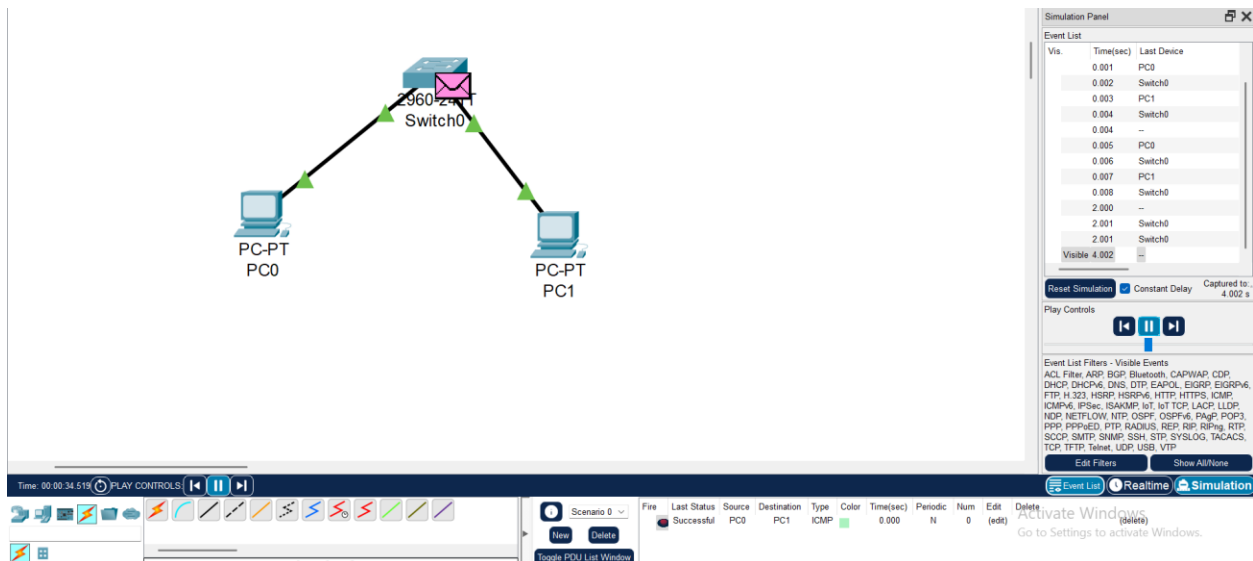
Observe Packet Flow (Optional, for learning):

- Click Simulation Mode at the bottom right.
- Click Add Simple PDU (envelope icon):
- Click on PC0 → then on PC1
- Run the simulation:
- Observe the ARP Request broadcast by PC0.
- PC1 responds with an ARP Reply.
- Switch learns the MAC addresses of both PCs.
- Data is forwarded based on MAC addresses.

Result:

- Successful communication was established between PC0 and PC1.
- ARP and MAC-based switching were correctly simulated.

- Ping test confirmed low-latency packet delivery.
- Switch forwarded frames intelligently without broadcasting after MAC learning.



2. Networking Between Two Computers (using Router)

Network Setup:

Open Cisco Packet Tracer.

From the device list, drag and drop:

- 2 PCs (PC0 and PC1)
- 1 router
- Use copper cross-through cables to connect:
 - PC0 to Switch0 (e.g., FastEthernet0/1)
 - PC1 to Switch0 (e.g., FastEthernet0/2)

Procedure

Step 1: Create the Topology

Place 2 PCs (PC0 and PC1) and 1 Router in the workspace.

Connect:

PC0 to Router's GigabitEthernet0/0

PC1 to Router's GigabitEthernet0/1

(Use Copper Straight-through cables for both connections)

Step 2: Assign IP Addresses to PCs

On PC0:

IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

On PC1:

IP Address: 192.168.2.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

Step 3: Configure Router Using GUI (No CLI)

Click on the Router.

Go to the Config tab.

a) Configure GigabitEthernet0/0:

Click on GigabitEthernet0/0 from the left menu.

Set:

IP Address: 192.168.1.1

Subnet Mask: 255.255.255.0

Check the box for Port Status: On (This acts as no shutdown).

b) Configure GigabitEthernet0/1:

Click on GigabitEthernet0/1.

Set:

IP Address: 192.168.2.1

Subnet Mask: 255.255.255.0

Enable the Port Status by checking the box.

Results

PC0 (192.168.1.2) successfully sent packets to PC1 (192.168.2.2) via the router.

No CLI commands were used; all configuration was done using the graphical interface.

